

Global Normalization in Joshi’s Arithmetic Teichmüller Theory: What the Identification Carries — and What Carries the Theorem

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Abstract

This assessment asks what the same global normalization carries in Theorem 6.10.1 of Joshi’s ATS IV (arXiv:2403.10430v2). Under the explicitly place-wise Artin–Whaples reading, number-field absolute values are identified definitionally; the rule selecting the coordinate λ_y remains an open correspondence question. An elementary product-formula lemma constrains multipliers of the standard valuations to one global scalar. The exclusion of a fixed product-formula hyperplane remains conditional on assumptions (F1)–(F2). The three normalization branches are diagnostic, not exhaustive. This corrective version distinguishes the source’s asserted Frobenius stability of a collated family from the unverified transport to the two indexed hulls in E2. It corrects three source readings: ATS IV states its height domain as $\overline{L}^*$, E2 equates absolute values of log-volumes, and the algebraic holomorphic hull of ATS III is not generically a convex closure. An explicit counterexample gives $\text{conv}(\mathbb{Z}_p \cdot 1) = \mathbb{Z}_p \cdot 1 \subsetneq \mathcal{O}_E = \text{hull}(\mathbb{Z}_p \cdot 1)$ in an unramified quadratic extension $E/\mathbb{Q}_p$. This does not refute the specialized tensor-region assertions or the ATS bounds. Normalized additive Haar measure is kept separate from powers of an absolute value: $m(p\mathbb{Z}_p) = p^{-1}$ does not become p^{-c} by changing a valuation exponent. Accordingly the previous positive claim of cross-frame measure pinning is withdrawn in every branch. A hull- and measure-compatible indexed transport is named as a possible sufficient route stronger than the displayed absolute-value equality, not as a proved or necessary condition. The selection rule, indexed E2 transport, compatible measure and weight bindings, and order comparison remain open. No proof or disproof of *abc*, Mochizuki’s Corollary 3.12, or Joshi’s estimates is claimed.

Scope and claim boundary.

- This paper proves *nothing* about the *abc* conjecture, about inter-universal Teichmüller theory, about Mochizuki’s Corollary 3.12, or about the correctness of Joshi’s bounds. It contains no proof and no disproof of any of these.
- It *verifies no bound*. No proof line of ATS IV Theorem 6.10.1, of Corollary 9.11.1.1, or of Corollary 3.12 was followed step by step; no estimate was recomputed.
- What it does: it *classifies definitions and the structure of justification* in one named theorem, and states for each of its components whether that component is definitional, supplied by construction, branch-conditional, or open.
- It takes no position in the dispute between Joshi and Scholze–Stix [9] on the mobility of the product-formula hyperplane. Where that dispute is touched, it is reported, not adjudicated.
- The structural analogy sketched in Section 8.2 is a working hypothesis of the present program about the *shape of arguments*. It is not a mathematical connection and, in particular, not a claim about the Riemann hypothesis.
- The central open item of this version — that the passages we read state no rule selecting the normalization coordinate of a moved arithmeticoid (Section 3.5) — is a *definitional gap*: a rule we did not find, which may well stand in material we have not read. It is not a refutation of ATS III Corollary 4.2.2.4 and not an error claim of any kind. Lemma 1 (Section 3.6) constrains which rules the product formula *alone* can supply; it does not assert that no such rule exists.
- Where a tension between two passages of a source is recorded, this is an invitation to resolve it and a candidate for correspondence with the author — not a claim that it is unresolvable, and not an error claim.
- **Textual basis** are `pdftotext` extracts of the arXiv PDFs. All line references (“extract l. NNNN”) are finding aids into those extracts and are *not* pagination of the published documents. Direct visual readback covered the source-PDF pages for equation (5.3.3), Theorem 5.10.1(3)–(4), Theorem 10.20.1(3), Definition 4.4.2/formula (4.4.5), and Theorem 6.10.1. No blanket OCR-robustness claim is made; in particular, the E1 absolute-value bars and the direction of the fiber exponent were checked in the source PDFs.

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1 Introduction: the point at issue

1.1 The finding that prompted this paper

The present paper is the fourth part of a small research program whose third part was a structured “ledger test” of the argumentative architecture of ATS IV: a requirement-by-requirement check of which load-bearing steps of the paper are supplied, which are referenced, and which are asserted. That test returned one requirement — the one concerning the comparability of the two arithmeticoids entering Theorem 6.10.1 — as met only *in part*. The reason was not a gap in a computation. It was that the identification carrying the middle equality of the theorem is, in the text’s own word, a tautology (ATS IV, extract l. 3272–3273), while the very same theorem’s remark forbids the place-by-place comparison that a reader might otherwise use to check it.

The two textual poles are these. On the one hand the theorem opens with a choice:

“Choose the same global normalization for the number field provided by the arithmeticoids y'_0 and y_0 . With this choice of simultaneous global normalization one has ...” (ATS IV, extract l. 3185–3187)

prepared in the introduction by

“Each arithmeticoid $y'_0 = \phi(y_0)$, y_0 provides an isomorphic copy of our number field and hence one has chosen the same global normalization for both the copies, the [sic] it does not matter which arithmeticoid is used for global computations.” (ATS IV, extract l. 740–743)

On the other hand, immediately below the theorem, the same text states:

“as arithmeticoids $y'_0 \neq y_0$ as they differ by Frobenius at each prime $v \in V_L$ and this, together with choice of concurrent global normalization for the copy of the (same) number field they provide *precludes direct comparison of various local quantities*.” (ATS IV, extract l. 3207–3211)

Both passages are Joshi’s. They motivate a question about the identification of number-field element scales, but do not place the entire E2 equality on that identification. The indexed sets, algebraic hulls, measures and weights require their own bindings.

Question 1. Does the phrase “same global normalization”, as used in ATS IV Theorem 6.10.1, *assert* an isometry between the two valuation scales, or does it *establish* one by definition?

The two answers have very different consequences. If the phrase asserts an isometry, then something is being claimed that a reader is entitled to see proved, and the program’s working name for the missing statement — a *global normalization contract lemma* — would be justified. If the phrase establishes the element-scale identity definitionally, then no separate lemma is needed for that limited identity. What carries the indexed hull and volume equality is a further question.

This paper answers Question 1 in the second sense — conditionally on a reading that the sources leave open, as Section 3.5 sets out — and then pursues the follow-up question, because the follow-up is where the substance turned out to be.

1.2 Sources, labels, and citation discipline

Four documents are used as primary text. Throughout we use the short labels in the left column; every source citation in this paper carries such a label, because the four extracts have

Table 1: Primary source corpus and assigned labels in this assessment.

Label	Source	Role in this paper
ATS II	arXiv:2303.01662v3 [2]	behavior of valuations along the Frobenius fiber
ATS II(1/2)	arXiv:2305.10398v12 [1]	definition of $\text{arith}(L)^{\text{nor}}$; period mapping; heights
ATS III	arXiv:2401.13508 [3]	normalization convention; hyperplane; volume theory
ATS IV	arXiv:2403.10430 [4]	Theorem 6.10.1 and its surrounding apparatus

independent line numbering and a bare line number would be ambiguous. Table 1 summarizes the primary source corpus and assigned roles.

In Joshi’s own bibliographies these appear as “[Joshi, 2023b]” (ATS II), “[Joshi, 2023a]” (ATS II(1/2)) and “[Joshi, 2024c]” (ATS III); the assignment of the first two was verified against the bibliography of ATS IV (ATS IV, extract l. 3807–3810), the third is the identification used consistently in the working notes underlying this paper. Where we reproduce a passage, the wording is verbatim from the extract; mathematical symbols have been typeset, and ellipses are marked. Two OCR-level readings were made explicitly: the string “ptvh-power” in ATS II(1/2) Remark 5.9.2 is read as “ p_v -th power”, and the subscript/superscript placements in the displayed normalization equations were checked against their surrounding prose.

1.3 What follows

Section 2 sharpens the question: the naive form “is F invariant?” misses the structure, because both the evaluated object and the scale in which it is evaluated move. Section 3 establishes the meaning of the phrase under the place-wise reading of the definition of $\text{arith}(L)^{\text{nor}}$; its Section 3.5 then states the trilemma that the definition leaves open, and Section 3.6 proves the lemma that constrains it. Section 4 shows by an elementary derivation, under two assumptions stated with it, that the product-formula hyperplane really does move, which excludes — under those assumptions — the one reading under which the phrase would have had non-trivial content. Section 5 classifies the terms of Theorem 6.10.1 and locates the candidate carrier. Section 6 separates local conventions from the open cross-frame measure and hull obligations. Section 7 states the open question at the margin — the one that the trilemma grew out of. Section 8 draws the consequences and states the limitations. Verdicts that depend on the branch are marked as such where they are stated; the summary of which verdict holds under which branch is Table 3.

2 Sharpening the question: equivariance, not invariance

2.1 Both sides move

A first attempt to formalize Question 1 would ask: is the quantity F invariant under a change of arithmeticoid? That formulation is wrong, and seeing why is the first substantive step.

In the absolute-value equality (with the indexing detailed in E2 below)

$$|\log \text{Vol}(\text{hull}(\tilde{\Theta}^{\varphi(y_0)}))| = |\log \text{Vol}(\text{hull}(\tilde{\Theta}^{y_0}))|$$

both arguments move: the evaluated object and the scale in which the evaluation happens. The correct form of the statement is therefore not invariance but an indexed, frame-dependent comparison. An isometric transport would be a possible sufficient route; the absolute-value equality alone does not imply it. This matters in both directions. A statement of this type is not automatically trivial — being an isometry is a substantive property — but it is also not automatically empty when it holds, unlike a mere renaming. Whether the phrase “identically normalized” asserts such an isometry or produces it by an identification is exactly Question 1.

The relevant structural facts are stated by Joshi himself. The schemes are isomorphic:

“the schemes underlying these holomorphoids are isomorphic to X/L ” (ATS IV, extract l. 426–427)

while the analytic structures, and with them the valuation scales, are not:

“arithmetic (and geometry) provided by the holomorphoids $\text{hol}(X/L)_y$ and $\text{hol}(X/L)_{y'}$... occur at intrinsically different valuation scales” (ATS IV, extract l. 450–453)

and this difference is not an inconvenience of the construction but its purpose:

“It is precisely the existence of such intrinsic differences in valuation scales which the averaging techniques ... exploit” (ATS IV, extract l. 457–459)

2.2 Two precision levels of the question

Two versions of the question can be stated, of unequal maturity.

(L-weak) The invariance class. Let N be a fixed choice of identical global normalization for the copies of L provided by y_0 and $\varphi(y_0)$. Characterize the class

$$G := \{ F : F(y_0) = F(\varphi(y_0)) \text{ is provable under } N \},$$

where F is evaluated covariantly (object and scale transported together) and “provable” explicitly *excludes* that the equality follows from the definition of N or from a notational convention.

A proof that $F \in G$ would have the shape of a chain of equalities carrying $F(\varphi(y_0))$ into $F(y_0)$ using only steps that either run over data demonstrably fixed by φ , or exploit a demonstrated compensation of two changes. The text supplies a formal model for the first kind: the proof of ATS IV Proposition 4.4.4 reduces $\log(q_M) = \log(q_{L_{\text{mod}}})$ to the degree formula $\sum_{w|v} e_{w|v} f_{w|v} = [M : L_{\text{mod}}]$ (ATS IV, extract l. 1866–1928). That is a genuine proof by classical means — but it establishes invariance under *field extension*, not under φ , and is therefore a template here, not an answer.

Several source passages delimit the proposed class. ATS IV reports that Tate-parameter absolute values rise under Frobenius (ATS IV, extract l. 567–569), that the normalized Tate-idele degree is non-constant on Y_L (ATS IV, extract l. 2055–2075), and that height is written on $\bar{L}^* \times Y_L$ (ATS IV, extract l. 550–559). ATS III reports movement of the product-formula hyperplane (ATS III, extract l. 518–522). These are source-reported examples and indications, not four independently proved failures of Frobenius invariance. A displayed domain alone proves neither non-constancy nor variation along a particular Frobenius orbit. The height statement and F-NEU-1 retain the separate domain and proof obligations of Section 7; D-1 retains (F1)–(F2).

(L-strong) Order compatibility after summation. Give a criterion under which global invariance yields *order compatibility* between two families of local contributions, although place-by-place comparability is excluded. The side condition is not invented; it is Joshi’s:

“one cannot claim that $z_p \leq z'_p$ holds for all primes p without violating the middle equality of (1.5.1)” (ATS IV, extract l. 817–819)

This second version is the less developed of the two: it names the structure but offers no candidate class of admissible local distributions. We record that as the largest substantive arrear of the formulation phase; it is not resolved in this paper.

2.3 A separating criterion

The classification used in Sections 5 and 6 rests on a criterion that we state as a hypothesis, not as a theorem.

Criterion K (candidate). A quantity is the more likely to be invariant in the load-bearing sense, the more completely it factors over data that φ demonstrably fixes (the underlying scheme, field degrees, residue field cardinalities), and the less so, the more directly it reads the *valuation scale* $|\cdot|_{K_{y_v}}$ of the arithmeticoid.

The criterion’s discriminating power can be read off the theorem itself, because Theorem 6.10.1 contains both types side by side: its first equality concerns q_ℓ , whose defining formula runs over $\text{ord}_w(q_w)$ and $\log w$, while its middle equality concerns $\log \text{Vol}$, which through $\text{LogVol}(\pi\mathcal{O}) = \log |\pi|$ reads the scale directly. Separating the two was the first result of the formulation phase; carrying the separation through is the business of Section 5. Table 2 summarizes the resulting taxonomy of invariance classes under an arithmeticoid shift.

Table 2: Taxonomy of invariance classes and the separating Criterion K under an arithmeticoid shift φ .

Class	Invariance Character	Factorization Basis	Exemplars in ATS IV	Gap Role
T	Trivial / definitional	Factors over φ -fixed scheme & degree data ($[M : \mathbb{Q}], e, f$)	$\text{ord}_w(q_w), \log w, q_\ell$ (§ 4.4)	Harmless
B	Bound-invariance (global)	Preserved under global summation / product formula	$\sum_v \log x _v = 0,$ $A' = CA$	Constraining
F	Fiber-equivariance	Transforms equivariantly along Frobenius fiber	Period map $\text{per}_{\text{arith}}, \tilde{\Theta}$	Structural
N	Non-invariance (untilt)	Directly evaluates untilt valuation scale $ \cdot _{K_{y_v}}$	Tate idele $\log(TI),$ Kummer map	Obstructing

3 The definition under the place-wise reading: $\text{arith}(L)^{\text{nor}}$ is unique per point

3.1 Three candidate readings (not exhaustive)

Before consulting the definition, the phrase “same global normalization” admits at least three readings. It is useful to state these three because two can be excluded on grounds internal to the text; the list is diagnostic, not a claim of logical exhaustiveness.

N1. Both arithmeticoids are normalized in the sense of the standing convention, i.e., the product formula holds in each.

N2. Both have the *same* product-formula hyperplane, $H_{y_0} = H_{\varphi(y_0)}$.

N3. A coordinated choice is made within the *residual freedom* that the normalization condition leaves open — the thought being that $\prod_v |x|_{K_v} = 1$ does not fix the individual $|\cdot|_{K_v}$.

For example, a further mechanism could supplement the product-formula condition with a global selection rule that fixes the residual scalar c , without returning to the place-wise Artin–Whaples binding of branch (a)/N1’ and without allowing the place-selective freedom of

branch (c)/N3. These three readings therefore organize the source passages at hand; they do not partition every possible selection rule.

Reading N1 cannot be what is meant on its own, because the convention is already universal in the series:

“I will also assume throughout this paper that all arithmeticoids $\text{arith}(L)_y$ are normalized i.e. $\text{arith}(L)_y = \text{arith}(L)_y^{\text{nor}}$... i.e. the valuations on each component of y are normalized so that the Artin–Whaples product formula ([10]) holds for each $x \in L^*$: $\prod_{v \in V_L} |x|_{K_v} = 1$ ”
(ATS III, extract l. 858–863)

Under N1, the sentence in Theorem 6.10.1 would add nothing that does not hold anyway. Reading N2 stands in visible tension with the text’s own emphasis on the mobility of the hyperplane, to which we return in Section 4. That leaves N3 as the reading that would carry content — and N3 is precisely the one whose definition is not stated in ATS III or ATS IV, but in the earlier paper.

3.2 What the definition actually says

The definition of the normalized arithmeticoid is given in ATS II(1/2) §5.3. The relevant passages are:

“The fundamental global constraint ... is the validity of the product formula for L ... for the *standard choice of normalizations* for valuations $(L_v, |\cdot|_v)$ (given in [Artin and Whaples, 1945]) ... (5.3.1) $\prod |x|_v = 1$ ”
(ATS II(1/2), extract l. 1840–1848)

“Now suppose one is given a $y = (\psi_v) \in Y_L$... Since the two valuations $|\cdot|_v$ and $|\cdot|_{K_{\psi_v}}$ on L_v are *equivalent*, *there exist, for each $v \in V_L$, real numbers $\lambda_v \in \mathbb{R}$ such that* (5.3.3) $(L_v, |\cdot|_v) = (L_v, |\cdot|_{K_{\psi_v}}^{\lambda_v})$.”
(ATS II(1/2), extract l. 1852–1858)

“(5.3.5) $\lambda_y = (\lambda_v)$... *is the normalization coordinate of y* . When valuations induced on $\{L_v\}$ by the local data of $\text{arith}(L)_y$ are normalized so that (5.3.4) holds, I will write this as $\text{arith}(L)_y^{\text{nor}}$ ”
(ATS II(1/2), extract l. 1867–1873)

“Because the absolute values $(|\cdot|_{K_{\psi_v}})$ depend on $y \in Y_L$, one has (5.3.7) $\lambda_{y'} \neq \lambda_y$ in general and hence one says that the *product formula normalization of y moves with $y \in Y_L$* . Especially, one sees that there is *no uniform choice* of the normalization coordinate λ_y for all $y \in Y_L$.”
(ATS II(1/2), extract l. 1884–1891)

Two further pieces of context belong here. Each point y carries a *canonical* valuation, obtained via the Fargues–Fontaine curve [12], and alongside it the L_v -*standard normalized* valuation with $|p|_{K_v} = 1/p^{f_v}$ (ATS II(1/2), extract l. 802–840). And the impossibility of normalizing everything at once is stated as a statement about the family:

“Regardless of the choice of a normalization ... one cannot normalize valuations of all untilts of $\bar{L}_v$ *simultaneously*. More precisely the function $y \mapsto |p|_{K_y}$ is not a constant function.”
(ATS II(1/2), extract l. 802–840, Rem. 2.7.1)

3.3 Three consequences

(i) Read as a place-wise binding, (5.3.3) determines the normalization uniquely for each fixed y . The exponent λ_v is fixed by (5.3.3) as the conversion exponent between the canonical K_{ψ_v} -valuation and the Artin–Whaples standard valuation. This is the classical uniqueness statement for equivalent absolute values: if $|\cdot|_1$ is non-trivial and $|\cdot|_2 = |\cdot|_1^a$ for some $a > 0$, then a is unique, being computable as $\log |y|_2 / \log |y|_1$ for any y with $|y|_1 > 1$ [11, Prop. 7.8, p. 108]. Read this way there is *no residual freedom* within which anything could be “coordinated”. That reading is natural — and it is the one this section’s verdict rests on — but it is not the only one the sources admit; Section 3.5 states the alternative.

(ii) **On elements of the number field the normalized valuations agree with the standard ones, for every y .** That is the content of (5.3.3)/(5.3.4). What moves with y is the *coordinate* λ_y — the position of the standard scale relative to the canonical untill scale — not the value $|x|$ for $x \in L$.

(iii) **The non-simultaneity is a statement about the family.** Remark 2.7.1 denies a y -uniform normalization across Y_L ; it is not an obstruction to the pointwise normalization of any individual y . Distinguishing the family statement from the point statement is what makes (i) and the quoted “no uniform choice” compatible rather than contradictory.

3.4 Decision of the trichotomy, under the place-wise reading

- **N3 is vacuous** — if (5.3.3) is read as a place-wise binding. Section §5.3 then leaves no residual freedom: the freedom conjectured in the formulation phase — that $\prod_v |x|_v = 1$ does not pin the individual absolute values — does not exist, because it is not *some* product-formula-compatible family that is chosen, but *the* one bound to the Artin–Whaples standard by (5.3.3). Section 3.5 shows that this qualification is not idle: elsewhere in the series a residual freedom of exactly the N3 kind is *exercised*.
- **N2 is excluded under the assumptions (F1)–(F2)** of the derivation of Section 4; without them it is textually disfavored but open. The exclusion is branch-independent — it uses only the movement of λ_y along a Frobenius fiber, which every branch retains — but it is not assumption-free: (F1)–(F2) place the standard point of ATS IV on such a fiber and supply the coordinate transformation at two finite places.
- **N1’ holds under that reading and is then the meaning of the phrase.** By N1’ we mean a sharpened form of N1: both copies are taken in their (individually unique) normalized form, whereby elements of the number field receive the same standard absolute values in both copies. This is the only thing that “Choose the same global normalization for the number field provided by the arithmeticoids y'_0 and y_0 ” can mean, given a place-wise reading of §5.3, and it is exactly what Joshi’s own justification says: the arithmeticoids “provide an isomorphic copy . . . hence one has chosen the same global normalization”.

The wording as a “choice” is, in this light, mildly misleading: there would be nothing to choose. It would be the invocation of the standing convention of ATS III, under which all arithmeticoids are normalized. Question 1 would thereby be answered in its second alternative — the phrase establishes the identity by definition and does not assert an isometry — and on the level of number-field elements a “missing contract lemma” of the conjectured form would not exist.

We write this in the conditional deliberately. The verdict is sound *given* the place-wise reading of (5.3.3), and that reading is the natural one for the wording of §5.3 taken by itself. It is not, however, the reading under which the normalization is used in the rest of the series. The next subsection sets out the resulting alternative, and every later verdict of this paper is marked for the branches under which it holds.

Independently of the branch, the definition identifies *nothing* on the untill level, where the canonical valuations of neighboring points on a Frobenius fiber demonstrably differ already on $\mathbb{Q}_p$. Which of the quantities in Theorem 6.10.1 live on which level is the subject of Section 5.

3.5 What the definition does not state: a trilemma for λ_y

The preceding subsection assumed an answer to a question the sources do not address: *which rule selects the normalization coordinate λ_y when the arithmeticoid moves?* We searched the

passages listed in Section 1.2 for such a rule and did not find one. What we did find is that the two candidate answers are both used — in different places.

The place-wise reading is what § 5.3 says. Equation (5.3.3) fixes λ_v as the exponent converting the canonical valuation into the Artin–Whaples standard one, and (5.3.5) speaks of *the* normalization coordinate. On that reading (5.3.4) is a rewriting of (5.3.1) rather than an additional constraint — the text itself says the product formula “reads” that way.

A place-selective freedom is what the later papers exercise. ATS III states that the scaling at the semi-stable places compels a re-normalization *at the other places*:

“the non-trivial scaling relationship established by Theorem 4.2.2.1(3) at primes $w \in V_{\text{odd},ss}$ forces that suitable (re)normalization of valuations must be introduced at primes $w \in V_{L'} - V_{\text{odd},ss}$ so that the product formula to continue to hold for each normalized arithmeticoid $\text{arith}(L)_{y_j}^{\text{nor}}$.” (ATS III, extract l. 1600–1609)

The same mechanism appears as a structural property in Theorem 4.6.1(4) (ATS III, extract l. 1786–1791), and ATS IV explains the movement of heights by it: the local contributions are “*bound together* . . . by means of the (global) product formula”, so that the global Frobenius “changes global heights in a subtle and complicated way in order to ensure that the product formula holds” (ATS IV, extract l. 585–589). An instruction to introduce a re-normalization at other places presupposes that at those places something is still free to be chosen — which is precisely the freedom that a place-wise reading of (5.3.3) denies.

Three diagnostic branches (not exhaustive). Since no rule is stated, three source-motivated possibilities must be kept apart:

- (a) **Standard-bound.** λ_y is the tuple determined place by place by (5.3.3). Then $\text{arith}(L)_y^{\text{nor}}$ is well defined, and Section 3.4 applies as written — but there is nothing to “introduce” at the other places, and the instruction quoted above has no object.
- (b) **Product-formula-bound.** λ_y is any tuple for which the product formula holds. Then it is well defined only up to a scalar, by Lemma 1 below, and the freedom is *not* place-selective.
- (c) **Place-selective.** λ_y is selected by some further rule that leaves genuine place-by-place freedom. This is what the passages just quoted use — but the rule itself is not given in the material we read, and it is not reconcilable with reading (5.3.3) as a place-wise binding.

Whether (a), (b) or (c) — or a further global selection rule such as the one noted above — is intended cannot, as far as we can see, be decided from the text. It is the natural first question of any correspondence about this material, and we formulate it as such in Section 7.

3.6 What the product formula alone can supply: a rigidity lemma

Branch (b) can be settled. The product formula constrains the multiplier tuple far more strongly than one might expect: it determines it completely, up to one global scalar.

Conversion multipliers used by the lemma. Let $\lambda_{y,v}^{\text{std}} > 0$ satisfy $|x|_{K_{y,v}}^{\lambda_{y,v}^{\text{std}}} = |x|_v^{\text{std}}$ on L_v , and write $\lambda_{y,v} = \mu_v \lambda_{y,v}^{\text{std}}$. Then $|x|_{K_{y,v}}^{\lambda_{y,v}} = (|x|_v^{\text{std}})^{\mu_v}$. The following lemma constrains μ , not directly λ_y . Branch (a) fixes $\mu_v = 1$; branch (b) restricts positive conversion exponents to $\mu_v = c > 0$. It makes no corresponding assertion about Haar measures.

Lemma 1 (Rigidity of the product-formula multiplier). *Let L be a number field, and for each place v let $|\cdot|_v^{\text{std}}$ denote the Artin–Whaples standard absolute value, so that $\sum_v \log |x|_v^{\text{std}} = 0$ for all $x \in L^*$. Let $(\mu_v)_{v \in V_L}$ be real numbers with*

$$\sum_{v \in V_L} \mu_v \cdot \log |x|_v^{\text{std}} = 0 \quad \text{for all } x \in L^*.$$

Then all μ_v are equal.

Proof sketch. For $L = \mathbb{Q}$ the statement is immediate and needs nothing beyond the standard absolute values. Substituting $x = p$ for a prime p gives $\log |p|_p = -\log p$, $\log |p|_\infty = +\log p$ and $\log |p|_q = 0$ for primes $q \neq p$, so the hypothesis reads $(\mu_\infty - \mu_p) \log p = 0$, i.e., $\mu_p = \mu_\infty$ for every prime p ; $x = -1$ adds no condition, and $\mathbb{Q}^*$ is generated by -1 and the primes.

For a general number field the same two steps are used, with two classical inputs. First, for a unit $x \in \mathcal{O}_L^*$ all finite terms vanish, and by Dirichlet’s unit theorem the vectors $(\log |x|_v)_v$ arch span the trace-zero hyperplane of the archimedean coordinates; hence μ_v is constant $= c$ over the archimedean places. Second, for a finite place v let h be the class number of L and π a generator of the principal ideal $\mathfrak{p}_v^h$; substituting $x = \pi$ gives $-\mu_v \cdot h \log N(\mathfrak{p}_v) + c \cdot h \log N(\mathfrak{p}_v) = 0$, i.e., $\mu_v = c$. $\square$

Status and provenance. The lemma is elementary and certainly not new in substance; we state it because the constraint it expresses is used implicitly in the material and, as far as we could determine, is nowhere cited there. The $\mathbb{Q}$ -case above is complete as it stands. The general case uses Dirichlet’s unit theorem and the finiteness of the class number, both classical; we give no section number for either, because we did not verify one in an edition we hold. We deliberately do *not* attribute the lemma to [10]: the product-formula characterization proved there concerns which fields admit such a system of absolute values, and we could not obtain the full text to check whether a uniqueness statement about the multipliers is included. Anyone citing this lemma should either prove it, as here, or verify a source directly.

Consequence for branch (b), stated with care. By Lemma 1 the admissible tuples in branch (b) are exactly $\mu \equiv c$ with $c > 0$. Two things follow, and they pull in opposite directions.

First, the freedom that branch (b) grants is *uniform*, never selective. A global factor multiplies every local contribution alike; it cannot raise the contribution at one place and lower it at another. Wherever the material describes a re-normalization at the non-semi-stable places as *compensating* a scaling at the semi-stable ones, that description asks for more than the product formula by itself can deliver. This is a statement about what one constraint implies — it is not a claim that Corollary 4.2.2.4 is false, and not a claim that no rule achieving the described effect exists. It says that such a rule would have to come from somewhere other than the product formula, and it thereby sharpens what a resolution of the trilemma would have to supply.

Second, the scalar c is not fixed by the constraint. For a projective object such as the hyperplane H_y this is harmless — c cancels, since coefficient vectors define the same hyperplane exactly when they are proportional — but for a quantity that is linear in λ , such as a height, it is not: the value changes by the factor c . Branch (b) therefore leaves the scale of such quantities undetermined unless a further convention is added.

Table 3: Which verdict of this paper holds under which branch of the trilemma of Section 3.5. “—” marks a verdict that the branch does not support as stated. Branch-independence is not the same as assumption-freedom: the first row holds in every branch *under the assumptions* (F1)–(F2) of Section 4, and the fifth row identifies the carrier of the middle equality without establishing the transport to the two indexed hull evaluations displayed in (E2) (Section 8.3).

Verdict	(a) standard-bound	(b) PF-bound	(c) place-selective	Evidence level
N2 excluded (D-1, under (F1)–(F2))	holds	holds	holds	O
N3 vacuous	holds	partly: freedom exists but is not selective	—	O
Scale identification definitional (N1')	holds	up to the scalar c	—	D
E1 (q_ℓ) invariant	holds	holds	holds (reads no scale)	D
E2 candidate carrier (φ -stability of $\tilde{\Theta}$; indexed transport untraced)	holds	holds	holds	S
E2 measure compatibility	open	open	open	O
Nothing identified on the untilt level	holds	holds	holds	D

D: definitional; **S**: source-asserted; **O**: open or conditional.

4 The period mapping really moves: the conditional exclusion of N2

4.1 The mobility of the hyperplane in the sources

Each normalized arithmeticoid provides a degree homomorphism and, through the logarithm of the product formula, a hyperplane:

“(2) Each normalized arithmeticoid $\text{arith}(L)_y$ provides a degree homomorphism $\dots (x_v) \mapsto \sum_{v \in V_L} \log(|x_v|_{K_{y_v}}) = \deg_{y_j}((x_v)_{v \in V_L})$. (3) As a consequence, each normalized $\text{arith}(L)_y^{\text{nor}}$ provides a hyperplane $H_{y_j} \subset V_L$ given by the logarithm of product formula equation (2.1.1)” (ATS III, extract l. 1762–1783)

The hyperplane is not fixed:

“the above arithmetic period mapping is a highly non-trivial global arithmetic period mapping in which an arithmetic avatar $\text{arith}(L)_y$ provides a hyperplane H_y (as above) *which moves under symmetries (such as the actions of the absolute Galois group, global Frobenius and L^*)*” (ATS III, extract l. 518–522)

and, correspondingly, “the global arithmetic period mapping is non-constant” (ATS III, extract l. 869–870). The reason given is the interplay of local freedom and global constraint:

“the valuations on untits parameterized by a Fargues–Fontaine curve at some arbitrary prime $\dots$ cannot be all simultaneously normalized and on the other hand, local changes of valuations are globally constrained by the validity of the product formula” (ATS III, extract l. 515–518)

In the earlier paper the hyperplane is given *explicitly with the normalization coordinates as coefficients*:

$$“H_y = \{z = (z_v) \in \bigoplus_v \mathbb{R} : \sum_v \lambda_v z_v = 0\}” \quad (\text{ATS II}(1/2), \text{ extract l. 2199–2206})$$

together with the statement that “one has a natural and *non-constant* mapping $y \mapsto H_y \dots$ called the period mapping” (ATS II(1/2), extract l. 2227–2233), whose non-constancy is proved as being “immediate from the fact that *normalizations of arithmeticoids do not carry over* because locally at each prime v , valuations *cannot be simultaneously normalized*” (ATS II(1/2), extract l. 2238–2243). Remark 5.10.2(5) of the same paper notes that the impossibility objection of Scholze–Stix [9] addresses exactly this phenomenon; ATS III likewise records that Scholze–Stix dispute the mobility of H_y (ATS III, extract l. 522–525). We report this disagreement and take no side in it.

4.2 The behavior along the Frobenius fiber

On the fiber $\{y_n\}_{n \in \mathbb{Z}}$ over the canonical point — the deck-transformation, i.e., Frobenius, orbit — the sources state:

- the Kummer embeddings $\mathbb{Z}_p(1) \rightarrow B$ attached to y_n and y_{n-1} “differ as $\mathbb{Z}_p$ -submodules by a p -multiple” (ATS II, extract l. 2119–2120);
- “One has $|p|_{K_{y_{n-1}}} = |p|_{K_{y_n}}^{1/p}$ ” (ATS II, extract l. 2122);
- “the valuations of p , and hence of the Tate parameters of the holomorphoids $\dots$ grow along the fiber $\{y_n\}$ ” (ATS II, extract l. 2127–2131);

- “each y_n comes with a canonical valuation on its residue field K_{y_n} ... but the valuations provided by y_{n-1} and y_n are not the same on $\mathbb{Q}_p$ ” (ATS II, extract l. 2184–2188).

and, for the action at all places at once, that the global Frobenius relates the local multiplicative structures of L and $L^{(1)}$ “through the p_v -th power mapping for each $v \in V_L$ ” (ATS II(1/2), extract l. 2149–2153).

4.3 The derivation

Conditional derivation D-1. Assume **(F1)** that the standard point y_0 used in ATS IV §4.1 lies in a Frobenius fiber to which ATS II Theorem 10.20.1(3) applies, and **(F2)** that ATS II(1/2) Remark 5.9.2 induces $\lambda_{\varphi(y),v} = p_v \lambda_{y,v}$ at two finite places of distinct residue characteristic. Then $H_{\varphi(y)} \neq H_y$: the product-formula hyperplane is not preserved by the global Frobenius.

Derivation.

1. H_y has coefficient vector λ_y (ATS II(1/2), extract l. 2199–2206).
2. Along the canonical fiber one has $|p|_{K_{y_{n-1}}} = |p|_{K_{y_n}}^{1/p}$ (ATS II, extract l. 2122). With the standard condition $|p|_{K_{y_n}}^{\lambda_n} = 1/p$ (case $L = \mathbb{Q}$, $v = p$) this yields $\lambda_{n-1} = p \cdot \lambda_n$.
3. At every place v , φ acts as a p_v -th power step (ATS II(1/2), extract l. 2149–2153), i.e., $\lambda_v \mapsto p_v \cdot \lambda_v$. The step from the quoted statement — which speaks of the local *multiplicative structures* — to this statement about the *normalization coordinates* is read off the quotation by analogy with step 2; it is carried out as a computation only in the case $L = \mathbb{Q}$, $v = p$ there. This is assumption (F2).
4. Two coefficient vectors define the same hyperplane only if they are proportional by a *single* scalar c . But $(p_v \lambda_v) = c(\lambda_v)$ would force $p_v = c$ for all v , which is impossible because the residue characteristics vary. Hence $H_{\varphi(y)} \neq H_y$. $\square$

Status of this derivation. D-1 is *our own* derivation — elementary algebra plus a proportionality argument — assembled from statements quoted verbatim above. It is *not* a theorem of the source, and it is not presented as one. It is stated here because it settles reading N2, which the formulation phase had been able to classify only as “documented-unlikely”.

Two caveats, carried explicitly.

- (i) Step 2 computes in the canonical fiber over the canonical point. That the standard points y_0 fixed in ATS IV §4.1 lie on such a fiber is consistent with the statement that “there are countably infinite possible choices for the standard point y_0 ... Any one of the countably many choices for y_0 are equally valid choices” (ATS III, extract l. 1723–1729), but it was not separately established here.
- (ii) The behavior at archimedean places is asserted in Remark 5.9.2 by the phrase “for each $v \in V_L$ ”, but was not separately checked. For D-1 this is not needed: two finite places with distinct residue characteristics already suffice for step 4.

Under these two caveats — which are the assumptions (F1)–(F2) of the statement above — N2 is excluded rather than merely improbable, and the fixed-point reservation that the formulation phase had to carry — the possibility that y_0 might happen to be a φ -fixed point — is relieved: along canonical fibers the normalization coordinate demonstrably moves, and a fixed standard point would have to lie outside such fibers, for which nothing in the material speaks.

5 Term classification of Theorem 6.10.1: three kinds of statement

5.1 The chain

In simplified notation, Theorem 6.10.1 asserts the chain

$$-|\log(q_\ell^{\varphi(y_0)})| = -|\log(q_\ell^{y_0})| \quad (\text{E1})$$

$$-|\log(q_\ell^{y_0})| \leq -\frac{1}{\ell^*} |\log \text{Vol}(\text{hull}(\tilde{\Theta}^{I_M, \varphi(y_0)}))| = -\frac{1}{\ell^*} |\log \text{Vol}(\text{hull}(\tilde{\Theta}^{I, y_0}))| \leq C_\Theta |\log(q_\ell^{y_0})| \quad (\text{E2})$$

(ATS IV, extract l. 3180–3205), where the first inequality is attributed to Mochizuki [7] via the corresponding corollary of ATS III. Our concern is the middle equality of (E2) and the equality (E1) — not the inequalities, which we do not examine.

5.2 E1: q_ℓ is a classical, y -free quantity

The q -divisor of ATS IV § 4.4 is defined by

“let q_w at C_{M_w} be a Tate parameter (*well defined up to multiplication by a local unit and its order at w*). . . . $q_M = \sum_{w \in V_M^{\text{odd}, ss}} \text{ord}_w(q_w) \cdot w$ ” (ATS IV, extract l. 1831–1846)

The definition chain used by the theorem can be displayed compactly as

$$q_M = \sum_{w \in V_M^{\text{odd}, ss}} \text{ord}_w(q_w) w, \quad \log(q_M) = \frac{1}{[M : \mathbb{Q}]} \deg(q_M) = \frac{1}{[M : \mathbb{Q}]} \sum_w \text{ord}_w(q_w) \log w,$$

followed, immediately before Theorem 6.10.1, by $q_\ell^{y_0} = \frac{1}{2\ell} \log(q)$ computed in the y_0 frame (ATS IV, extract l. 1831–1876, 2523–2528, 3180–3189). Here q is the § 5.4 shorthand for $q_{C/M}$, hence for the relevant § 4.4 divisor q_M ; correspondingly, $\log(q)$ denotes the normalized arithmetic degree $\log(q_M)$, not a new object. Thus the superscript records the computational frame; the displayed scalar factors through the classical divisor and degree data.

so that $\text{ord}_w(q_w)$ is the classical discrete-valuation order in the local field M_w ; the parenthesis declares it a well-defined invariant of the definition. The proof of Proposition 4.4.4 is correspondingly classical throughout: uniformisers $\pi_w, \pi_v, q_v = \pi_v^\alpha$, $\text{ord}_w(q_w) = e_{w|v} \text{ord}_v(q_v)$, $\log w = f_{w|v} \log v$ (ATS IV, extract l. 1852–1876). *No y , no untilt and no holomorphoid occurs in it.* The q -divisor is a y -free classical object of the pair (C_M, M) .

This settles E1 *independently of the trilemma*, and the point is worth making explicitly, because it is the one verdict here that needs no branch. The quantities entering q_ℓ are integer orders $\text{ord}_w(q_w)$ and residue-field cardinalities. Neither reads an absolute value supplied by an arithmeticoid, and re-scaling a valuation by a positive exponent leaves the associated order unchanged — a re-normalization in any of the three senses of Section 3.5 therefore cannot move q_ℓ . So whichever branch is intended, $q_\ell^{y_0}$ and $q_\ell^{\varphi(y_0)}$ compute the *same classical number*; the superscript records the ambient computational frame, but does not change the value. Under branch (a) the same conclusion can be reached via N1' — both normalized copies read number-field elements in the same standard scale — but that route is not needed, and using it as the reason would make an unconditional verdict look conditional. Direct visual readback of the ATS IV source PDF, p. 66, also shows absolute-value bars on *both* sides of E1. The one-sided version in the text extract was an extraction artefact. Hence no additional assumption such as $\log(q_\ell) \geq 0$ is required: E1 is definitionally true under the stated identification of q_ℓ with the § 4.4 q -divisor.

5.3 The apparent contradiction, and its resolution

At first sight this collides with an explicit statement of the same paper: “the absolute values of Tate parameters rise under global Frobenius ϕ ” (ATS IV, extract l. 567–569). If Tate parameters rise, how can a quantity built from them be invariant?

The resolution is that two *different objects* are involved. First, the classical q -divisor of § 4.4, which is a combination of ord_w -values and is y -free — this is what E1 speaks about. Second, the Tate idele and the untilt Tate parameters, which are z -dependent; these are what § 1.4(2), Proposition 4.5.12 and the fiber statements of ATS II speak about:

“Let $\text{arith}(L)_z^{\text{nor}}$ be the *normalized* arithmeticoid ... the normalized arithmetic degree $\log(TI_{C/\text{arith}(L)_z})$ is a function of z ... one obtains a *non-constant* function ‘normalized degree of the Tate divisor’ $Y_L \rightarrow \mathbb{R}$ ” (ATS IV, extract l. 2055–2075)

Note that this non-constancy is asserted *for the normalized* arithmeticoid: normalization does not remove it, because it does not live on the number-field level at all.

The same distinction disposes of a trap that the formulation phase had marked as open. Joshi does document a holomorphoid dependence — “an important point of the theory is that $\text{Tate}(C/M)$ depends on the choice of a holomorphoid $\text{hol}(C/L)_y$ and one should write $\text{Tate}(C/M)_y$ to indicate this choice” (ATS IV, extract l. 681–684) — but he documents it for $\text{Tate}(C/M)$, and he introduces q precisely because Tate lacks a stability property:

“The divisors $\text{Tate}(C/M)$, $\text{Tate}_2(C/M)$ are not stable under passage to field extensions ... so one replaces $\text{Tate}(C/M)_2$ by the normalized arithmetic degree of another divisor $q = q(C/M)$ (§ 4.4), which is similar to the divisor Tate, but which receives contributions only from primes of M lying over a prime ... of semi-stable reduction ... By Proposition 4.4.4, this new divisor q is stable under base-field extensions.” (ATS IV, extract l. 698–705)

Transferring the documented y -dependence of Tate to q would be a fallacy of object substitution; the index sets alone differ. What resolves the matter is not a compensation mechanism but the observation that the y -variance lives in a different formalization.

5.4 E2: family stability and the open indexed equality

The source PDF (ATS IV v2, p. 66) equates *absolute values* of log-volumes. The former transcription without those bars asserted a stronger statement. From $|a| = |b|$ one cannot infer $a = b$ without a common sign binding; no such binding has been established here for the exact indexed hulls and volume conventions in E2.

The source assertion: stability of the collated set. ATS IV describes an adelic set which collates tuples of Tate parameters from holomorphoids and says it is stable under iterates of the global Frobenius (ATS IV, extract l. 506–515). This is evidence level S, a source assertion. It identifies a candidate family-level carrier; it does not identify the $I_M, \varphi(y_0)$ and I, y_0 subsets, their ambient algebras, or their measures. We have not traced those bindings.

Named transport: a sufficient stronger route, still open. Write $T : B_{I_M, \varphi(y_0)} \longrightarrow B_{I, y_0}$ for a proposed map. One possible sufficient route would establish the ambient algebra and integral structures, an isometry, the set identity $T(\tilde{\Theta}^{I_M, \varphi(y_0)}) = \tilde{\Theta}^{I, y_0}$, hull compatibility $T(\text{hull}(S)) = \text{hull}(T(S))$, and preservation of the relevant measures and degree weights. These conditions would yield equality of the corresponding log-volumes, hence their absolute values. They are stronger than the source’s absolute-value equality; they have not been shown necessary. The map and each compatibility remain open. No replacement measure proof is claimed from the generic hull counterexample or from a special algebra decomposition.

The scale identification is only one layer. Under branch (a), number-field elements have the standard absolute values in each copy. Under branch (b), the product formula restricts the conversion multipliers to a global scalar, but does not bind that scalar across frames. Neither observation proves E2 or transports normalized Haar measure. Under branch (c), even the element-scale identification requires its own selection rule. The measure obligations of Section 6 are open in all three branches.

5.5 Where the work sits: the asymmetry of the two bounds

A third textual observation completes the picture. The two bounds of the chain are not computed in the same frame:

“by the virtue of stability under the global Frobenius morphism, one has the freedom of the choice of an arithmeticoid in the computing the *lower bound* ... [it] is *independent of the choice* ... the lower bound will be computed using $y'_0 = \varphi(y_0)$ while the upper bound will be computed using y_0 (*the upper bound is not independent of this choice*).” (ATS IV, extract l. 733–742)

The source also notes that a prior release omitted this Frobenius shift (ATS IV, extract l. 755). This locates a candidate role for the choice of frame; it does not settle the equality. Branch (a) makes the element-scale identification definitional, while the shift, indexed sets, hulls and measures still need compatible bindings. Branch (c) additionally leaves the selection-rule question. The present assessment does not verify either inequality or certify that family stability supplies the missing indexed step.

5.6 Classification table

Table 4 records the classification. Two rows are deliberately not forced into one of the invariance classes: $\tilde{\Theta}$ as a set is not a scalar-valued quantity to be classified as invariant or not, and $\log \text{Vol}(\text{hull}(\tilde{\Theta}))$ is the composite whose analysis is the subject of Section 6.

Criterion K has illustrative examples in the classical q -divisor and the untilt-sensitive terms. These examples do not make it a general invariance theorem and do not classify the algebraic hull or the weighted cross-frame log-volume as transport-verified.

6 Measure conventions and open cross-frame obligations

Absolute value, Haar measure and multiplication module. For each place and frame, distinguish the absolute value $|\cdot|_{y,v}$, the additive Haar measure $m_{y,v}$ normalized by $m_{y,v}(\mathcal{O}_{L'_w}) = 1$, and its multiplication module

$$\Delta_{y,v}(\alpha) = \frac{m_{y,v}(\alpha A)}{m_{y,v}(A)}$$

for a measurable set A of positive finite measure and $\alpha \neq 0$. The equality $\Delta_{y,v}(\alpha) = |\alpha|_{y,v}$ requires a compatible absolute-value convention. On $\mathbb{Q}_p$, normalization forces

$$m(\mathbb{Z}_p) = 1, \quad m(p\mathbb{Z}_p) = p^{-1}, \quad |p|_p^c = p^{-c}.$$

The second identity follows by partitioning $\mathbb{Z}_p$ into its p cosets modulo $p\mathbb{Z}_p$. Changing a valuation exponent does not change this normalized additive Haar measure. The scalar in Lemma 1 therefore provides no measure-transfer proof.

Table 4: Term classification of Theorem 6.10.1 with assigned Evidence Levels. Class T: definitional/trivially invariant (factors over φ -fixed data). Class N: demonstrably not invariant (reads until data). Evidence levels: **D** (definitional), **S** (source-asserted), **T** (transport-verified), **O** (open or conditional). The classification is independent of the trilemma of Section 3.5; in the last row, indexed transport and cross-frame measure compatibility remain open in every branch (Table 3). Class T and evidence level T are distinct labels.

Term	Factors over	Class	Evidence level	Behavior
$\text{ord}_w(q_w)$ (§ 4.4)	discrete-valuation structure of M_w (φ -fixed)	T	D	invariant
$\log w, [M : \mathbb{Q}], e_{w v}, f_{w v}$	residue and degree data (φ -fixed)	T	D	invariant
$q_\ell = \frac{1}{2\ell} \log(q)$	only the above	T	D	E1 definitional ; source PDF has absolute values on both sides
Tate idele degree $\log(TI_{C/\text{arith}(L)_z})$	untilt valuations	N	S	non-constant (Prop. 4.5.12)
Kummer embedding $\mathbb{Z}_p(1) \rightarrow B$	choice of untilt y_n	N	S	differs by a p -multiple (Thm. 10.20.1(2))
$\tilde{\Theta}_{\text{Mochizuki}}$ as a set	the whole family (collated)	—	S	φ -stable by construction
$\log \text{Vol}(\text{hull}(\tilde{\Theta}))$	set + measure convention	E2	O	Absolute-value E2: indexed transport and measure compatibility open in every branch

Local volume conventions and degree weights. ATS III defines normalized Haar measure on the number-field completions by $\text{Vol}_{L_w}(\mathcal{O}_{L_w}) = 1$ (ATS III, extract l. 6562–6572) and relates suitable fractional-ideal coset volumes to local absolute values (ATS III, extract l. 6599–6620, 6663–6676). It specifies reciprocal local degrees as weights (ATS III, extract l. 6678–6688). These are statements about specified fields, integral structures and measure conventions. Their numerical ingredients are classical; they do not identify the ambient tensor algebras, exact theta sets, hulls, weighted components or measures across E2 frames. Standard element scales under branch (a), or a common valuation exponent under branch (b), are insufficient. Branch (c) additionally leaves the element-scale choice open.

Algebraic hull need not equal convex closure. ATS III v4, §§ 9.10.7–9.10.8 (pp. 125–127), defines the holomorphic hull through integer rings of field components of the tensor algebra. The former generic identification with convex closure was incorrect. Take $E/\mathbb{Q}_p$ unramified quadratic, $E_1 = E$, $E_2 = \mathbb{Q}_p$, the identity algebra map, and $U = \mathbb{Z}_p \cdot 1$. The set is compact and already $\mathbb{Z}_p$ -convex. The smallest hull of the prescribed form $\alpha\mathcal{O}_E$ containing 1 is $\mathcal{O}_E$: if $1 \in \alpha\mathcal{O}_E$, then $\alpha\mathcal{O}_E \supseteq \mathcal{O}_E$, with equality attained for $\alpha = 1$. Thus

$$\text{conv}(U) = \mathbb{Z}_p \cdot 1 \subsetneq \mathcal{O}_E = \text{hull}(U).$$

Strictness follows from the $\mathbb{Z}_p$ -ranks one and two. Convexity of a hull containing U gives only $\text{conv}(U) \subseteq \text{hull}(U)$ in general. Admissibility of this U as a specialized Mochizuki tensor product region has not been established. The example refutes our generic identification, not Proposition 9.10.8.1(3), Theorem 9.11.1, or *abc*. A proposed replacement through decomposition of finite étale algebras still needs the actual algebra maps, integral structures, sets, weights and indexed transport; it is not a verified positive measure argument.

Revised conclusion. The former assertion that the measure side factors without remainder over number-field data, or is pinned under branch (a) and up to a scalar under branch (b), is withdrawn. Cross-frame measure compatibility remains open in every branch. Identifying element scales does not close indexed E2.

Table 5: Components after correction. D: definitional; S: source-asserted; O: open or conditional.

Component	Basis	Level	Status
Element scales	Place-wise standard reading	O	Definitional under (a); global scalar under (b); open selection under (c)
E1	Classical q -divisor	D	Under the stated identification; absolute values on both sides
Collated family	Frobenius-stability assertion	S	Candidate carrier; indexed E2 not established
Indexed E2	Absolute values of log-volumes	O	Open in every branch; unsigned equality not inferred
Measure and hull	Haar modules, integral structures, algebraic hull, degree weights	O	Compatibility open in every branch; generic hull equality withdrawn
Selection rule	Coordinate λ_y	O	Open correspondence question
Local comparison	ATS IV Remark 6.10.2	S	Expressly precluded

7 Where the trilemma comes from: the height passage

The trilemma of Section 3.5 was not part of the question this paper set out to answer. It emerged from a passage encountered while reading, and this section records how, because the guiding principle of the program is to document what was seen rather than to defend a position.

Finding F-NEU-1. The height passages of ATS II(1/2) §§ 8.2–8.3 are indicators of a possible tension under the place-wise reading. The claimed non-constancy, its proof using a local valuation, and its domain $\mathbb{P}^n(\overline{L})$ must be checked separately. A selection rule for λ_y may clarify the reading, but this paper has not settled the domain or proof questions. F-NEU-1 remains open.

The definition uses the *normalized* valuations explicitly: “Choose the standard normalization ... work with $\text{arith}(L)_y^{\text{nor}}$... work with the normalization coordinate λ_y ”, giving $h(P) = \sum_v \max_j \log(|x_j|_{K_{v_v}}^{\lambda_v})$ (ATS II(1/2), extract l. 2947–2960); and Lemma 8.2.3 derives the independence of the height from the choice of homogeneous coordinates precisely from the fact “that we work with a normalized arithmeticoid” (ATS II(1/2), extract l. 2982–2986). Under a place-wise reading of (5.3.3) these normalized values agree, on elements of L_v , with the standard ones — for *every* y — so for $P \in \mathbb{P}^n(L)$ the height would be the classical, y -independent height in the sense of [13]. ATS IV says as much about the definition: “except for explicating the dependence on the arithmeticoid, the definition of height ... is the same as the one given in [Bombieri and Gubler, 2006, § 1.5.1]” (ATS IV, extract l. 548–550).

Proposition 8.3.1, however, asserts that “ $h_{\text{arith}(L)}$ depends *non-trivially* on $\text{arith}(L)$ ” (ATS II(1/2), extract l. 2990–3004), and its proof invokes the fact that “ $y \mapsto |p|_{K_y}$ is a non-constant function” — which is the *canonical*, unnormalized valuation function of Remark 2.7.1. Remark 8.3.2 adds that “the normalization of $\text{arith}(L)_{y_1}^{\text{nor}}$ *cannot be carried over* to

$\text{arith}(L)_{y_2}^{\text{nor}}$ (ATS II(1/2), extract l. 3008–3015), which closes the one easy escape — keeping λ_{y_1} fixed while y moves.

Three possible readings require separate checks. The definition in ATS II(1/2) § 8.2 and Lemma 8.2.3 uses a normalized height; its relationship to canonical local valuations must not be settled by replacing one object with another. In particular, Proposition 8.3.1 states non-constancy of that normalized height on $\mathbb{P}^n(\bar{L})$, although its proof also uses an unnormalized local absolute value. The statement, proof step and domain remain distinct review obligations.

The proposed restriction of y -dependence to $\bar{L} \setminus L$ cannot be rejected merely from ATS IV:545: the original PDF v2, p. 13, says $x \in \bar{L}^*$, not $x \in L^*$. Its application to integer abc -triples is a separate passage (ATS IV, extract l. 608–616). Independently, ATS II(1/2) v12, Theorem 8.7.1 (pp. 54–55), really uses $x \in L^*$ and asserts general strictness (ATS II(1/2), extract l. 3210–3216). This is a distinct textual indication, not a substitute for the incorrect ATS IV quotation. The family assertion that normalization cannot be carried over does not by itself produce strictness at a fixed point. F-NEU-1 therefore remains an open correspondence question; none of these observations establishes a global error in ATS.

What the passage does supply is the mechanism, and it is the mechanism that generates the trilemma. In § 8.5 the movement of the height is derived from the product-formula constraint:

“since one is working with normalized arithmeticoids $\text{arith}(L)^{\text{nor}}$, the normalization constraint i.e., the product formula (5.3.4) implies that *changes in p -adic valuations forced by the action of n must be compensated by changes at other primes* so that the normalization constraint ... continues to hold.”
(ATS II(1/2), extract l. 3162–3193)

Compensation at other places presupposes freedom at those places. So the operative meaning of $\text{arith}(L)^{\text{nor}}$ in this passage is not “bound place by place to the Artin–Whaples standard” but “constrained, globally, by the product formula” — and then the question is which rule picks the coordinate out of the admissible set. That is branch (b) or (c) of Section 3.5, and Lemma 1 says that (b) does not deliver what the passage needs, since the freedom the product formula leaves is a single global scalar and cannot be place-selective.

The resulting question, and its status. The precise question we would put to the author is therefore this: *which rule selects the normalization coordinate λ_y for a moved arithmeticoid?* Concretely — is the normalized arithmeticoid the one whose induced absolute values on L_v are the Artin–Whaples standard ones, as (5.3.3) suggests, or is $(\cdot)^{\text{nor}}$ a constraint on a choice, for which the product formula alone is not sufficient to pin down λ_y ?

We make no error claim, against Proposition 8.3.1 or against anything else. The resolving passage may well stand in material we have not read, or follow from a reading of the definition that the author would supply at once. The two textual observations above — the place-wise appearance of (5.3.3), and the place-selective use of the normalization in § 8.5, in ATS III Corollary 4.2.2.4 and in § 1.5 of ATS IV — are reported as observations. What has changed relative to the first version of this assessment is only this: the tension cannot be set aside as marginal, because the verdicts of Sections 3.4–6 depend on how it is resolved. That dependence is what Table 3 records. The reading of the sources underlying this section has been carried out in more detail in the companion verification of this program (Parts V and VI, in preparation).

8 Assessment and consequences

8.1 For the ledger method

The original ledger located a question at the phrase declaring the normalization identification a tautology. Under the place-wise reading, that phrase indeed identifies number-field element scales definitionally. This limited conclusion does not discharge E2: the source’s stability

assertion concerns a collated family, whereas the theorem evaluates two indexed algebraic hulls with weighted volume conventions.

The corrected audit separates these duties. It reports the family assertion at evidence level S and keeps the indexed equality, sign binding and cross-frame measure compatibility at level O. The generic hull identification used in the earlier positive measure argument is withdrawn. Under branch (c), selection of the normalization coordinate is a further open duty. A formally acceptable ledger or a source-citation check is not a proof of those duties and does not strengthen an unverified bound.

The method’s useful output here is the correction of the actual source readings and a more precise set of open obligations. It is not a completed bridge certificate or acceptance of the entire ATS argument.

Table 6: Selected Gap-Series papers (non-exhaustive crosswalk): Target claims, primary diagnostic instruments, and core normalization/transport verdicts.

Part / Paper	Target Claim	Primary Instrument	Core Verdict	Status
Part I (Method)	Inter-universal transport	Bridge-Contract Criterion (C1–C7)	Foundational methodology; specifies explicit verification duties	Published
Part IV (This Paper)	Global normalization in ATS IV Thm 6.10.1	Invariance classification & Trilemma taxonomy	Identification is definitional under (a); transport of $\tilde{\Theta}$ -hull untraced	Corrective draft v1.2
Part IV-B (BrXi App.)	Self-application of $\text{Br}\Xi$	Sub-bridge ledger audit	Scope restriction; prevents cyclical domain inheritance across boundaries	Draft v0.1
Part V (Substance)	Normalization weights in ATS	All-place rigidity classification	Excludes non-trivial place-selective scaling under global Artin–Whaples	Draft v0.1
Part VI (Repair)	Repair program for ATS chain	Effective-weight test & Case taxonomy	Proves all-place rigidity for product-formula weights; tests height chain	Draft v0.2

8.2 For global-only mechanisms: a cautious outlook

The structural pattern of Theorem 6.10.1 — “no comparison place by place, pinned globally” — is not peculiar to this construction. Three carriers of that pattern appear in the material, two of them from Joshi’s own text.

The prototype is the product formula itself [10]. The identity $\prod_v |x|_{K_v} = 1$ says nothing whatever about the individual $|x|_v$, and is nevertheless the load-bearing global constraint of the entire construction. The second carrier is Joshi’s own second example within the same proof passage: “ $A' = C \cdot A$. *This is a global equation, which only appears after one sums over all primes*” (ATS IV, extract l. 834–836). That two such statements sit next to each other in one argument suggests a recurring pattern rather than an idiosyncrasy of the middle equality. The third is the pattern examined in this paper, now available as a worked specimen: the hyperplane construction plus Remark 6.10.2 give an instance in which the global pinning and the local incomparability are both explicit and both sourced.

A working hypothesis of the present program — recorded in its own candidate register, and stated here with the caution it requires — is that such statements live on the collated family rather than at the individual place, and that this gives them a structural silhouette in common with statements available only after summation over all places, Weil positivity among them. We state this as a hypothesis about the *shape of arguments*. It is emphatically *not* a mathematical connection, and *not* a claim about the Riemann hypothesis or about any

consequence of one. The corresponding work package carries an explicit abort clause: should the collected carriers turn out to share nothing beyond the observation itself, the analogy is to be recorded as a negative result and the register entry downgraded — not quietly carried forward. A reverse-direction question in the vicinity of [14] is noted in the project’s concept document and is expressly not taken up here.

8.3 Source corrections and open obligations

Three corrected readings. The source-PDF domain in ATS IV v2, p. 13, is $\overline{L}^*$. ATS II(1/2) Theorem 8.7.1 separately uses L^* ; the two anchors are not interchangeable (Section 7). ATS IV v2, p. 66, places absolute-value bars around both log-volume terms in E2. No common sign of the exact indexed evaluations is proved here, so the former unsigned equality is withdrawn. ATS III v4, pp. 125–127, uses an algebraic hull; its generic equality with convex closure is refuted by the explicit example in Section 6. This does not decide the special tensor-region claims. The earlier E1 sign discrepancy remains separately resolved as an extraction artefact: the source has absolute-value bars on both E1 sides.

Open indexed equality and measure transfer. Family stability is source-asserted. A map between the exact indexed ambient algebras, its integral structures, theta-set images, hull action, measures and weights has not been traced. The named transport is only a sufficient stronger route to E2; it is not a necessary condition derived from the source. Neither a norm power nor a generic hull identity supplies the absent measure argument. These obligations remain open in every branch.

Selection and order questions. The rule selecting λ_y remains open. Lemma 1 constrains only multipliers of standard valuations satisfying the product formula; it supplies no place-selective rule or Haar-measure transport. D-1 remains conditional on (F1)–(F2). F-NEU-1 and the strong order question L-strong remain open. Separate follow-up audits are needed before positive E2 or measure statements from companion Parts V and VI can be imported as verified results.

8.4 Limitations

This subsection states what the paper does not do, in the terms in which the underlying working notes stated it.

1. **No proof was followed.** Neither Theorem 6.10.1 of ATS IV, nor the corollary of ATS III on which its first inequality relies, nor Mochizuki’s Corollary 3.12 [7] was verified line by line. No estimate was recomputed and no bound was checked. What was classified is the *terms and the structure of justification*, not the correctness of the inequalities.
2. **D-1 is our own derivation and is conditional**, not a theorem of the source: elementary algebra plus a proportionality argument, assembled from statements quoted verbatim. It holds under the assumptions (F1)–(F2) stated with it — the canonical-fiber assumption for the standard point, and the transfer of the coordinate transformation to two finite places, which is read off the quotation rather than computed there. The archimedean case is likewise unchecked and, for D-1, not needed.
3. **Element-scale verdicts are branch-conditional.** Definitional identification and the vacuity of N3 apply under branch (a); branch (b) leaves a global conversion scalar and branch (c) its selection rule. Cross-frame measure pinning is not proved under any branch and its former positive classification is withdrawn. D-1 requires (F1)–(F2); E1

retains its stated identification; family stability is source-asserted, not an indexed hull or volume proof.

4. **The trilemma is a definitional gap, not an error claim.** We did not find a selection rule for λ_y in the passages we read; we do not claim that none exists, and we make no claim about the correctness of ATS III Corollary 4.2.2.4 or of anything resting on it.
5. **Lemma 1 is stated with a deliberate reference restriction.** The $\mathbb{Q}$ -case is proved here in full; the general case invokes Dirichlet’s unit theorem and the finiteness of the class number as classical results without a section number, because we verified none. In particular we do not attribute the lemma to [10], whose full text we could not obtain.
6. **F-NEU-1 remains open.** The corrected ATS IV domain $\overline{L}^*$ and the separate ATS II(1/2) statement on L^* do not adjudicate the entire height argument or establish an error in ATS.
7. **The classification of E1 as definitional** holds modulo the reading that q_ℓ in Theorem 6.10.1 is the § 4.4 q -divisor. The definition given immediately before the theorem supports that reading, but the identification is ours. The source PDF itself displays absolute-value bars on both sides of E1.
8. **The measure analysis classifies conventions, not estimates.** The correctness of the volume estimate itself — the product bound, the ℓ^* -factors, the translation to Mochizuki’s formulation — was not examined.
9. **The strong form of the question is undeveloped.** The order compatibility question (L-strong) of Section 2 names a structure but no candidate class. This is the largest substantive arrear.
10. **The invariance catalogue is a starting inventory**, not a completeness claim. Arakelov and adelic degrees in particular remain unfilled, because the material examined says nothing usable about them and no reference was guessed.
11. **Textual basis and visual readback.** Quotations use `pdftotext` extracts; line references are finding aids, not pagination. The source-PDF pages for equation (5.3.3), Theorem 5.10.1(3)–(4), Theorem 10.20.1(3), Definition 4.4.2/formula (4.4.5), and Theorem 6.10.1 were visually checked. In particular, exponent direction and the two pairs of E1 absolute-value bars are load-bearing. No blanket OCR-robustness claim is made for other passages.
12. **No claim upgrade.** Nothing in this paper is a statement about the correctness of Joshi’s *abc* argument, about Mochizuki’s theory [5, 6, 7, 8], about the objections of [9], or about the Riemann hypothesis. It is the clarification of one definitional question and of its consequence for one conjectured lemma.

AI Disclosure

AI systems were used extensively in all phases of this work: conception and research discussion, literature and source work, mathematical analysis and proof development, creation and checking of the computational scripts, execution and evaluation of the computational runs, text, translation, and typesetting.

Quality assurance: use of several independent models from different providers with counter-checks and adversarial AI reviews; traceable, versioned computational scripts with stored

result files and checksums; continuously maintained proof and register documents; strict claim hygiene (assertions only with documented evidence, addenda instead of silent corrections); changelogs across all versions.

Restriction for this paper: this work involved no computational scripts and no computational runs; the corresponding items of the standard text (creation and checking of the computational scripts, execution and evaluation of the computational runs, versioned scripts with stored result files and checksums) do not apply here.

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